Supplementary B — Full mathematical derivations

Companion to: *A Band-Mask Robustness Diagnostic for LDA on HSI*

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A.1 Scope

This supplement expands Section II of the conference paper. We provide (a) the full generative model of LDA on HSI documents, (b) the variational lower bound minimised by online VB, (c) the analytic form of the chance-corrected adjusted Rand index used as paired-ARI, (d) the bipartite-matching formulation of the Hungarian alignment, and (e) the full pseudocode of the diagnostic.

B.2 LDA on band-frequency HSI documents

B.2.1 Tokenisation

Let $\mathbf{x}_d \in \mathbb{R}_+^B$ be a pixel spectrum on B bands. The canonical V1 tokenisation min-max-normalises across bands and quantises each band reflectance into $Q + 1 = 13$ levels:

$$\tilde{x}_{d,b} = \frac{x_{d,b} - \min_{b'} x_{d,b'}}{\max_{b'} x_{d,b'} - \min_{b'} x_{d,b'}} \in [0, 1], \tag{1}$$

$$c_{d,b} = \lfloor Q \cdot \tilde{x}_{d,b} \rfloor \in \{0, 1, \ldots, Q\}. \tag{2}$$

The document d is represented as the integer count vector $\mathbf{c}_d = (c_{d,0}, \ldots, c_{d,B-1})$, treating each band index b as a vocabulary item and $c_{d,b}$ as the number of times the band-token appeared in the document. The token vocabulary has size $V = B$ (one token per band) and the document length is $N_d = \sum_b c_{d,b}$.

B.2.2 Generative model

The standard LDA generative process [1] applies verbatim to the band-frequency corpus. For K topics and a vocabulary of size V, with hyperparameters $\alpha > 0, \eta > 0$:

$$\phi_k \sim \text{Dirichlet}(\eta \mathbf{1}_V), \quad k = 1, \ldots, K, \tag{3}$$

$$\theta_d \sim \text{Dirichlet}(\alpha \mathbf{1}_K), \quad d = 1, \ldots, D, \tag{4}$$

$$z_{d,n} \sim \text{Categorical}(\theta_d), \quad n = 1, \ldots, N_d, \tag{5}$$

$$w_{d,n} \sim \text{Categorical}(\phi_{z_{d,n}}). \tag{6}$$

The joint posterior $p(\theta, \phi, z \mid w, \alpha, \eta)$ is intractable; we approximate it with the online variational Bayes inference of Hoffman, Blei, and Bach [2].

B.2.3 Online VB lower bound

Let $q(\theta_d, z_d) = q(\theta_d \mid \gamma_d) \prod_n q(z_{d,n} \mid \lambda_{d,n})$ be the document-local variational distribution and $q(\phi_k) = q(\phi_k \mid \beta_k)$ the topic-word distribution. The evidence lower bound (ELBO) on document d is

$$\begin{aligned}\mathcal{L}_d(\gamma_d, \lambda_d, \beta) = \mathbb{E}_q[\log p(w_d, z_d, \theta_d, \phi)] \\ - \mathbb{E}_q[\log q(z_d, \theta_d, \phi)],\end{aligned} \tag{7}$$

which after expansion and use of the digamma identity $\mathbb{E}_{\text{Dirichlet}}[\log \pi_k] = \psi(\alpha_k) - \psi(\sum_{k'} \alpha_{k'})$ takes the form

$$\begin{aligned}\mathcal{L}_d = \sum_{n=1}^{N_d} \sum_{k=1}^{K} \lambda_{d,n,k} \big[&\psi(\gamma_{d,k}) - \psi(\textstyle\sum_{k'} \gamma_{d,k'}) \\ &+ \psi(\beta_{k,w_{d,n}}) - \psi(\textstyle\sum_v \beta_{k,v}) \big] \\ &- \sum_{n,k} \lambda_{d,n,k} \log \lambda_{d,n,k} + C(\alpha, \eta, \gamma_d, \beta). \end{aligned} \tag{8}$$

Maximisation w.r.t. λ yields

$$\lambda^*_{d,n,k} \propto \exp\big[\psi(\gamma_{d,k}) + \psi(\beta_{k,w_{d,n}}) - \psi(\textstyle\sum_v \beta_{k,v})\big]\,. \tag{9}$$

Maximisation w.r.t. γ yields

$$\gamma^*_{d,k} = \alpha + \sum_n \lambda^*_{d,n,k}. \tag{10}$$

The global β is updated via natural-gradient ascent on the stochastic mini-batch ELBO with the learning-rate schedule $\rho_t = (\tau_0 + t)^{-\kappa}$, $\tau_0 = 10, \kappa = 0.7$, mini-batch size 128, $T_{\max} = 60$ passes over the corpus.

B.2.4 Posterior summaries used in the diagnostic

After convergence we extract:

$$\phi_{k,b} = \frac{\beta_{k,b}}{\sum_v \beta_{k,v}}, \qquad \text{topic-word distributions,} \tag{11}$$

$$\theta_{d,k} = \frac{\gamma_{d,k}}{\sum_{k'} \gamma_{d,k'}}, \qquad \text{per-document topic posterior,} \tag{12}$$

$$z_d = \arg\max_k \theta_{d,k}, \qquad \text{dominant topic per document.} \tag{13}$$

C.3 Band-mask sweep

C.3.1 Mask specification

A band mask is a function $m : \{0, \ldots, B-1\} \to \{0, 1\}$ with $m(b) = 1$ if band b is retained. The masked document-term matrix restricts the count vector accordingly: $\mathbf{c}_d^{(m)} = (c_{d,b})_{b:m(b)=1}$. The masked vocabulary size is $V^{(m)} = |\{b : m(b) = 1\}|$.

The four canonical masks of the diagnostic are:

$$m_{\text{vnir}}(b) = \mathbf{1}[\lambda_b \in [400, 1100]\,\text{nm}], \tag{14}$$

$$m_{\text{swir}}(b) = \mathbf{1}[\lambda_b \in [1100, 2500]\,\text{nm}], \tag{15}$$

$$m_{\text{nw}}(b) = 1 - \mathbf{1}[\lambda_b \in W], \tag{16}$$

$$m_{\text{f50}}(b) = \mathbf{1}[b \in \text{TopK}_{50}(F)], \tag{17}$$

where $W = [1350, 1430] \cup [1800, 1950] \cup [2480, 2500]$ nm and F_b is the per-band Fisher discriminant ratio

$$F_b = \frac{\sum_\ell n_\ell (\mu_{b,\ell} - \mu_b)^2}{\sum_\ell n_\ell \sigma^2_{b,\ell}} \tag{18}$$

with $\mu_{b,\ell}, \sigma_{b,\ell}$ the per-class mean and standard deviation of reflectance at band b for class ℓ, computed on the *unmasked* spectrum.

C.3.2 Masked refit

For each m we refit the LDA model with the same $K, \alpha, \eta, T_{\max}$, seed as the canonical fit but using the restricted document-term matrix. This yields $\phi^{(m)}, \theta^{(m)}$, and a masked dominant-topic map $z_d^{(m)}$.

D.4 Paired adjusted Rand index

Definition D.4.1 (Adjusted Rand index)**.** Let $U = \{U_1, \ldots, U_K\}$ and $V = \{V_1, \ldots, V_{K'}\}$ be two partitions of the same set $\mathcal{S}$. Let $n_{kj} = |U_k \cap V_j|$, $n_k = |U_k|$, $n_j = |V_j|$, $n = |\mathcal{S}|$. The Rand index is $R = (a+b)/\binom{n}{2}$ where a counts pairs in the same cluster under both U and V and b counts pairs in different clusters under both. The adjusted Rand index corrects for chance via the standard form

$$\mathrm{ARI}(U, V) = \frac{\sum_{k,j}\binom{n_{kj}}{2} - \left[\sum_k \binom{n_k}{2}\sum_j \binom{n_j}{2}\right]/\binom{n}{2}}{\frac{1}{2}\left[\sum_k \binom{n_k}{2} + \sum_j \binom{n_j}{2}\right] - \left[\sum_k \binom{n_k}{2}\sum_j \binom{n_j}{2}\right]/\binom{n}{2}}.$$

The paired-ARI in the diagnostic is

$$\mathrm{ARI}_{\text{paired}}(m) = \mathrm{ARI}\Big(\{z_d\}_{d\in\mathcal{S}}, \{z_d^{(m)}\}_{d\in\mathcal{S}}\Big), \tag{19}$$

with $\mathcal{S}$ the set of pixels labelled in both the canonical and masked dominant-topic maps (sentinel 255 excluded).

E.5 Hungarian topic-id alignment

ARI is invariant under topic-id permutations of V, so it summarises how much agreement exists but not which masked topic corresponds to which canonical topic. We recover the correspondence by solving the maximum-weight bipartite matching on the confusion matrix.

Definition E.5.1 (Confusion matrix)**.** $C \in \mathbb{Z}_{\geq 0}^{K \times K^{(m)}}$ with $C_{k,j} = |\{d \in \mathcal{S} : z_d = k \wedge z_d^{(m)} = j\}|$.

Definition E.5.2 (Hungarian permutation)**.** $\sigma^* : \{0, \ldots, K-1\} \to \{0, \ldots, K^{(m)}-1\}$ is the solution of the linear-assignment problem

$$\sigma^* = \arg\max_{\sigma} \sum_{k=0}^{K-1} C_{k,\sigma(k)}, \tag{20}$$

computed in $O(K^3)$ time by the Hungarian algorithm [3], as implemented in `scipy.optimize.linear_sum_assign` [4, 5].

The swap rate under alignment is

$$\text{swap\_rate}(m) = \frac{1}{|\mathcal{S}|}\sum_{d\in\mathcal{S}} \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)], \tag{21}$$

the fraction of pixels whose dominant topic still changes after the masked topic ids have been relabelled to maximise diagonal mass.

Proposition E.5.3 (Relationship between paired ARI and swap rate)**.** *Under the canonical $K = K^{(m)}$ assumption, a paired ARI of 1 corresponds to swap rate 0, and vice versa. For $\mathrm{ARI}_{paired}(m) < 1$ the swap rate strictly bounds the disagreement in pixel labels under the best permutation; ARI in addition adjusts for the chance baseline given the marginal cluster sizes.*

Algorithm 1 Band-mask robustness diagnostic

Require: Pixel spectra $\mathbf{X} \in \mathbb{R}^{D\times B}$, ground-truth labels $\mathbf{y} \in \mathbb{Z}^{D}$, wavelengths $\boldsymbol{\lambda} \in \mathbb{R}^{B}$.

$\mathbf{C} \leftarrow \text{Tokenise}(\mathbf{X}; Q = 12)$ ▷ $D \times B$ counts
$\theta, \phi \leftarrow \text{OnlineVB-LDA}(\mathbf{C}; K, \alpha, \eta, T_{\max}, \text{seed})$
$z_d \leftarrow \arg\max_k \theta_{d,k}$ for $d = 1, \ldots, D$ ▷ canonical map
Compute Fisher rank F via Eq. 18.
for $m \in \{\text{vnir}, \text{swir}, \text{nw}, \text{f50}\}$ **do**
    $\mathbf{C}^{(m)} \leftarrow$ retain columns of $\mathbf{C}$ satisfying $m(b) = 1$
    if $V^{(m)} < 8$ **then**
        continue ▷ auto-skip Pavia U × SWIR
    end if
    $\theta^{(m)}, \phi^{(m)} \leftarrow \text{OnlineVB-LDA}(\mathbf{C}^{(m)}; K, \alpha, \eta, T_{\max}, \text{seed})$
    $z_d^{(m)} \leftarrow \arg\max_k \theta_{d,k}^{(m)}$
    $\mathcal{S} \leftarrow \{d : z_d \neq 255 \wedge z_d^{(m)} \neq 255\}$
    $\text{ARI}_{\text{p}}(m) \leftarrow \text{ARI}(\{z_d\}_{\mathcal{S}}, \{z_d^{(m)}\}_{\mathcal{S}})$
    Build $C \in \mathbb{Z}^{K\times K}$ from $\mathcal{S}$.
    $\sigma^*(m) \leftarrow \textsc{LinearSumAssignment}(-C)$
    $\text{swap}(m) \leftarrow |\mathcal{S}|^{-1} \sum_{\mathcal{S}} \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)]$
end for
return $\{\text{ARI}_{\text{p}}(m), \sigma^*(m), \text{swap}(m)\}_m$

F.6 Diagnostic pseudocode

G.7 Numerical complexity

Per (scene, mask) tuple: one OnlineVB-LDA fit at $O(T_{\max} \cdot D \cdot V^{(m)} \cdot K)$, one $K \times K^{(m)}$ confusion-matrix construction at $O(|\mathcal{S}|)$, one Hungarian assignment at $O(K^3)$. On the largest labelled scene (Salinas, $D = 3520, K = 12, V = 204$, $T_{\max} = 60$) the canonical-vs-masked diagnostic completes in ~ 12 seconds on a single CPU core.

References

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